Concealing Prices: How Delayed Price Disclosure Influences Consumer Purchase Decisions

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This article presents the first systematic empirical investigation into a longstanding question in retail: is it better to display prices upfront or reveal them later in the purchase process? Two large-scale field studies demonstrate that delayed price disclosure can either increase or decrease sales. Supporting laboratory studies reveal that one plausible explanation is that a price delay allows price beliefs to shift consumers' internal reference prices upward or downward, creating either positive or negative price expectation disconfirmations when prices are revealed. When consumers anticipate prices should be expensive (e.g., from premium brands or upscale stores), a price delay allows price beliefs to shift price expectations upward, making purchases more likely when prices are revealed. Conversely, when consumers anticipate prices should be inexpensive (e.g., sales events or discount stores), a price delay allows price beliefs to shift price expectations downward, making purchases less likely when prices are revealed. Our findings offer retailers actionable insights on when to reveal prices to customers. In doing so, the authors contribute to the literature on price obfuscation and challenge the conventional wisdom that shopping experiences should always minimize friction.

Keywords: Price disclosure, price obfuscation, consumer expectations, pricing, retailing

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A classic retail tactic is to withhold prices from consumers. Window displays have long invited passersby to "ask inside for prices." Print catalogs prompted readers to "contact us for prices." More recently, e-commerce websites urge shoppers to "add to cart to see price." Despite its persistence, businesses continue to debate whether "obfuscating" prices—defined as "actions that firms take to make price search more costly" (Ellison 2016, 288)—enhances or harms sales performance. Some proponents argue that "it is downright stupid to show your prices" alongside products (Silva 2018), claiming consumers must first internalize a product's value before being exposed to its cost. Conversely, opponents insist that retailers should be forthcoming and "own" the price conversation (Hubspot 2017), reasoning that delaying price disclosure (PD) creates confusion, eroding trust and drawing unwanted attention to cost.

Prior research has found that obfuscating a part of the price can be beneficial for retailers, especially when enough consumers in a market are unwilling or unable to determine what they should pay. For instance, advertising a low base price while obscuring ancillary charges (e.g., delivery costs, service fees, taxes) can lead to increased sales compared to disclosing the total cost upfront (Chiles 2021). This occurs because some consumers commit to the purchase before seeing the additional charges, whereas others disregard the charges once revealed. However, academic research has largely overlooked the impact of concealing the entire price of a product.

We propose that delaying PD influences sales by amplifying the impact of price beliefs on expected prices. Consumers hold pre-existing price beliefs based on product type (e.g., discount, luxury), quality level (e.g., low, high), retailer (e.g., discount, premium), or the economic environment (e.g., high inflation). When the disclosure of a price is delayed, consumers rely more heavily on these pre-existing beliefs to form price expectations. Consequently, expensive-price beliefs (inexpensive-price beliefs) shift price expectations upward (downward), making an offer price (OP) seem more (less) appealing once the price is revealed, thus increasing (decreasing) the likelihood of purchase. Consistent with these hypotheses, we show that delayed (vs. immediate) PD increases purchase intentions when consumers hold expensive-price beliefs. We demonstrate this effect in a field study at the online store of a mainstream retailer (study 1) and in laboratory experiments in which the product is premium (study 2) or sold by a premium retailer (study 5). We also show that delayed (vs. immediate) PD decreases purchase intentions when consumers hold inexpensive-price beliefs. We demonstrate this effect in a field study investigating the sales promotions of a mainstream retailer (study 3) and in laboratory experiments in which the product is on deal (study 4) or sold by a discount retailer (study 5). We also test a practical boundary condition of our effects: the disclosure of a minimum advertised price (MAP) policy as a reason for delaying the price, which moderates the effect of a delayed price (study 6).

Our research offers three key contributions. First, our research offers retailers insights into optimal PD timing, moving beyond the reliance on anecdotes and post hoc justifications that have previously dominated the debate. Second, it extends the price obfuscation literature by examining the understudied tactic of full price concealment and revealing its unique effect on shifting price expectations. Contrary to the common belief that obfuscation is universally beneficial, our findings demonstrate predictable situations in which it is detrimental. Finally, our findings engage with the broader management conversation about friction in the customer journey. Contrary to the prevailing advice to pursue "zero-friction" experiences—as exemplified by the vision for retail by Facebook (2023)—our studies show that introducing specific forms of friction (delaying PD) can be positive.

PRICE OBFUSCATION

Obfuscating Ancillary Charges

Price obfuscation has been studied through both theoretical and empirical research. Theoretically, hidden surcharges can be used to exploit "vulnerable" consumers who may intentionally or unintentionally underestimate the actual cost of a product. This exploitation arises from several factors: consumers within a market exhibit varying degrees of sensitivity to price differences (Ellison 2005), some demonstrate either myopia or sophistication regarding future expenses (Carlin 2009; Gabaix and Laibson 2006), individual differences in search costs exist (Ellison and Wolitzky 2012), and some consumers may simply be confused by obfuscation (Chioveanu and Zhou 2013). Despite the differing approaches to modeling consumer behavior, these studies consistently conclude that firms benefit from concealing prices.

Similarly, empirical studies on price obfuscation have consistently shown that "dripping" ancillary charges later in the purchase process, rather than disclosing them upfront, increases demand (Blake et al. 2021; Brown, Hossain, and Morgan 2010; Chetty, Looney, and Kroft 2009; Goldin and Homonoff 2013; Hossain and Morgan 2006; Morwitz, Greenleaf, and Johnson 1998). Specifically, in online shopping contexts, research indicates that starting eBay auctions with low opening prices and high shipping fees generates more bidders and higher revenue compared to bundling the base price and shipping fees (Hossain and Morgan 2006). The authors suggest that obfuscation exploits consumers' tendency to downplay surcharges. Further, eBay sellers have been shown to capture

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more revenue by shrouding shipping fees but not when they are disclosed upfront (Brown et al. 2010). Similarly, a field experiment on StubHub, a secondary marketplace for event tickets, showed that consumers were more likely to select and purchase more expensive tickets when mandatory fees were initially hidden (Blake et al. 2021). Analysis of click-stream data by the same authors also indicated that price shrouding hindered price comparisons.

Beyond the digital world, in physical retail settings, research indicates that disclosing a product's sales tax on the shelf, rather than at the cashier, reduces sales by approximately 8% (Chetty et al. 2009). This reduction was attributed to increased salience of the sales tax rather than consumer ignorance about taxes. This finding has been replicated and further explored, revealing that low-income consumers pay closer attention to taxes levied at the register compared to high-income consumers (Goldin and Homonoff 2013). In laboratory settings, studies have shown that separating a buyer's premium from the product price increases demand (Morwitz et al. 1998). This is partly due to consumers anchoring on the base price and insufficiently adjusting for the premium when calculating the total price.[1]

Obfuscating an Entire Price

To our knowledge, there are only two articles that study the obfuscation of an entire price. First, Lynch and Ariely (2000) manipulated price "usability" such that participants in the high usability condition saw prices upfront and could sort products according to this information, whereas those in the low usability condition saw prices at a later stage (and, therefore, had no sorting option). They found that "increasing price usability had no reliable effect" on price sensitivity (Lynch and Ariely 2000, 100), but this conclusion relied on one laboratory experiment with a sample of 72 participants for an eight-cell between-subjects design. Second, Karmarkar, Shiv, and Knutson (2015) used functional magnetic resonance imaging (fMRI) to study the relationship between product primacy (product information before price information) or price primacy (price information before product information) and brain activity. Although the authors found that product primacy shifted the focus of participants from assessing monetary worth to assessing product liking, the effect on purchase intentions was unclear. In addition, although price primacy has been demonstrated in laboratory settings, it is uncommon in real-world contexts—unlike the obfuscation of an entire price, which is frequently observed in practice.

Research Opportunity

In summary, the price obfuscation debate—whether businesses should reveal prices upfront or later in the purchase process—spans various sectors and retail formats, generates conflicting opinions, yet it lacks substantial empirical evidence. The existing literature on price obfuscation highlights two key avenues for new research. First, whereas all obfuscation strategies inherently delay price discovery, the practice of delaying the disclosure of an entire price is sufficiently common to merit focused investigation. Second, if delayed PD indeed alters purchase behavior, current explanations within the price obfuscation literature may be insufficient. When businesses withhold the full price, it is unlikely that consumers precommit to a purchase or misrepresent price information once revealed. Therefore, a deeper examination of this tactic is warranted.

PRICE EXPECTATIONS

Price expectations, also known as internal reference prices (IRPs), serve as benchmarks against which consumers evaluate OPs (Kalwani et al. 1990; Monroe 1973; Winer 1986). These expectations are shaped by three primary factors: (1) knowledge of the product's price history, (2) product attributes, and (3) price beliefs. Knowledge about the price history of the product helps a consumer understand the value a seller, and the marketplace, assign to a product as well as price trends (Kalyanaram and Winer 1995; Monroe 1973; Winer 1986). Product attribute levels help a consumer form price expectations when price history is unavailable (Shirai and Meyer 1997). For example, a first-time home buyer knows that the neighborhood, square footage, and quality of internal features (e.g., flooring, cabinets, fixtures) influence price. Price beliefs influence the construction of, or adjustments to, price expectations (Hamilton 2024). In this discussion, we focus on how price beliefs influence price expectations.

Price Beliefs

Price beliefs are typically directional in that a consumer can believe a product is going to be inexpensive or expensive. There are three primary sources of price beliefs. First, there are beliefs about a retailer's price image (Hamilton and Chernev 2013). Consumers can use the retailer's nonprice characteristics (e.g., ambiance, décor, and location) and pricing policy (e.g., price advertising, price points, promotional history) to form beliefs that the retailer has inexpensive (e.g., Walmart) or expensive (e.g., Nordstrom) prices.

Second, there are price beliefs about categories or subcategories of products (Cheng and Monroe 2013; Mazumdar, Raj, and Sinha 2005). For example, premium product categories (e.g., diamonds, caviar, yachts) should be more

1 There is far less research on the long-run effect of price obfuscation. In one article, Santana, Dallas, and Morwitz (2020) find that self-justification and mistaken perceptions about additional search costs, or potential gains from switching sellers, made initial decisions excessively sticky. Yet, Chiles (2021) argues that reputational concerns counteract the incentive to companies to obfuscate prices especially if consumers can punish deceptive behavior. Specifically, Chiles (2021) finds that shrouding resort fees lowers guest ratings in the US hotel industry.

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expensive than value product categories (e.g., ramen noodles, dish rags, cotton swabs). Likewise, premium products in a category (e.g., designer purse, haute couture) should be more expensive than basic products in the same product category (e.g., affordable handbag, fast fashion). Subcategory price beliefs are based on differences in attributes or attribute levels between subcategories. For example, espresso machines with programable settings (vs. not) belong to the premium subcategory. Similarly, espresso machines with a dual-boiler system, versus a single-boiler system, belong to the premium category. Consumers aggregate product features and represent beliefs at the subcategory level because it is more cognitively efficient, though it is possible for consumers to use beliefs about individual attributes to inform price beliefs.

Third, there are nonattribute events that are used to make inferences about a product's price (e.g., advertising, external reference prices, general economic conditions; Biswas and Blair 1991; Cheng and Monroe 2013). Regardless of the source, price beliefs can be used to construct price expectations when limited information is available or to adjust price expectations that are initially based on product attributes. If price beliefs are shared by consumers in a target market, these beliefs will shift the distribution of price expectations for the target market (figure 1; Hamilton and Chernev 2013).

Manufacturers and retailers can proactively influence price beliefs and alter consumers' expectations about price, independent of the product itself. For example, premium retailers design their spaces (a store's layout, fixtures, window displays, etc.) to convey an image of exclusivity, luxury, or meticulous quality. These actions can prompt shoppers to anticipate even higher prices (i.e., distribution D in figure 1), which will result in more unit sales at a current OP or similar sales at a higher OP (Vigneron and Johnson 1999; Wiedmann, Hennings, and Siebels 2009). Similarly, the product mix offered by a retailer plays a crucial role in creating and reinforcing price beliefs. An assortment dominated by premium products signals that the retailer caters predominantly to an upmarket clientele, reinforcing the expectation of higher prices (Allard and Griffin 2017; Simonson 1999).

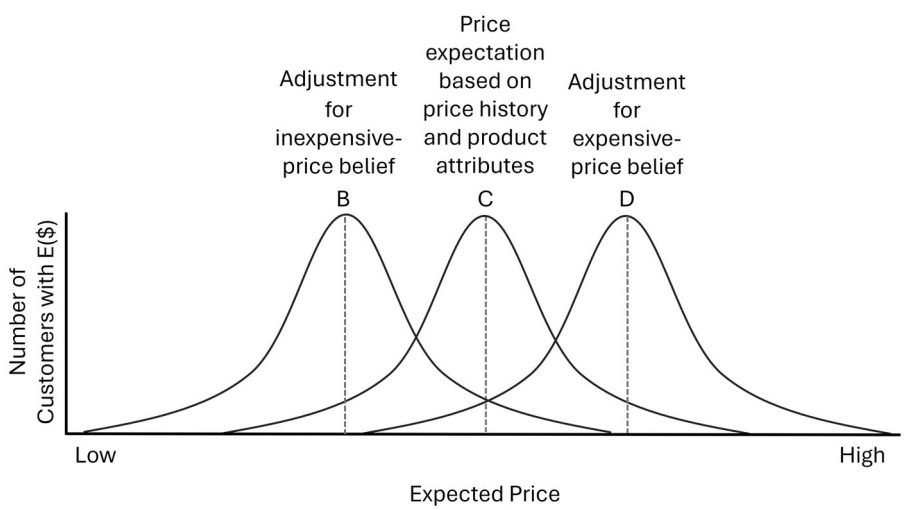
Discount retailers are positioned to encourage consumers to believe they offer lower prices (i.e., distribution B in figure 1). Discount retailers design their spaces to convey an image of affordability, frugality, or essential quality to attract customers who desire or require low price points (Chang and Wang 2014; Grace and O'Cass 2005). Similarly, low-end retailers offer product assortments comprised primarily of "value" or "economy" alternatives to signal that they cater to a budget-conscious clientele (Simonson 1999). These actions encourage and reinforce inexpensive-price beliefs.

Price Delays, Price Beliefs, Price Expectations, and Purchase Behavior

Price expectations are a function of a weighted average of a product's price history, product attributes, and price

FIGURE 1

DISTRIBUTIONS OF CUSTOMER PRICE EXPECTATIONS



NOTE.— Curve C illustrates the baseline distribution of a target market's price expectations, derived from price history and product attributes. Curve B (D) represents adjustments to price expectations owing to the target market's belief that prices will be inexpensive (expensive). E($), expected price.

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beliefs. Price beliefs have a stronger influence on price expectations when there is uncertainty about how to assign value to product attributes, such as when there is a limited history of purchasing and/or using a product (Shirai and Meyer 1997). Price beliefs can influence the anticipated performance of individual attributes (Plassmann et al. 2008; Rao 2005) or exert a general, halo-like influence on the expected price (Hamilton 2024). Price beliefs can also exert a stronger influence on price expectations when contextual factors make these beliefs more salient, as is the case when a price is absent (i.e., delayed) and consumers must infer pricing information.

When the price is delayed, as opposed to disclosed immediately, price beliefs can exert a greater influence in shaping price expectations. In other words, a customer's belief about whether a price will be inexpensive or expensive has a significantly larger impact on their price expectations when the price is delayed rather than disclosed immediately. This occurs because the absence of price information creates a knowledge gap that consumers fill by relying on their preexisting beliefs about what the price should be.

The predicted influence of price beliefs and price delays on price expectations is illustrated in figure 2. The top panel shows a situation in which the belief is that prices will be expensive. This belief shifts the distribution of consumers' price expectations from distribution C (i.e., baseline price expectation based on price history and product attributes) to distribution D (i.e., adjustment for expensive-price belief). A price delay amplifies the relative weight of the expensive-price belief in the price expectation judgment, further shifting consumer price expectations to distribution E (i.e., amplification of the expensive-price belief). The bottom panel of figure 2 shows the opposite effect. When the belief is that prices will be inexpensive, the distribution of consumers' price expectations shifts from distribution C (i.e., baseline price expectation based on price history and product attributes) to distribution B (i.e., adjustment for inexpensive-price belief). A price delay amplifies the relative weight of the inexpensive-price belief in the price expectation judgment so that the price expectation shifts from distribution B to distribution A (i.e., amplification of the inexpensive-price belief).

Shifts in price expectations should influence consumer purchase behavior and, by extension, the overall sales in a target market. To illustrate, consider the top panel of figure 2. When an expensive-price belief shifts consumers' price expectations from distribution C to distribution D, more consumers will have expectations exceeding the OP. When considering the OP, these consumers should experience a favorable price expectation disconfirmation (i.e., [mean expected price of all target market consumers in the presence of expensive-price beliefs minus offer price] is positive) because the OP is below their expected price (Kalyanaram and Winer 1995). Consumers with a favorable price expectation disconfirmation should be more willing to purchase the product because the OP is better than expected. A price delay, which further shifts price expectations to distribution E, should increase the number of consumers with a favorable expectation disconfirmation even further, leading to higher sales. Of course, this prediction assumes homogeneity in consumer price beliefs—specifically, that all consumers in the target market share expensive-price beliefs, as consumers with inexpensive price beliefs would shift their price expectations downward.

Similar predictions can be made in the lower panel of figure 2: inexpensive-price beliefs should shift price expectations downward (distribution B), and even more so with a price delay (distribution A). This results in fewer consumers willing to purchase the product, as more of them will experience an unfavorable price expectation disconfirmation. Again, heterogeneity in price beliefs will weaken this effect.

The hypotheses represented in figure 2 are summarized below.

H1: When consumers believe a price will be expensive, a delayed price (compared to an immediate price):

a. will increase purchase intention and/or sales;
b. because of a more favorable price expectation disconfirmation (mediator).

H2: When consumers believe a price will be inexpensive, a delayed price (compared to an immediate price):

a. will decrease purchase intention and/or sales;
b. because of a more unfavorable price expectation disconfirmation (mediator).

OVERVIEW OF STUDIES

We test the hypotheses as follows. Study 1 is a field study that tests hypothesis 1a (i.e., a price delay increases sales) at the online store of a mainstream consumer retailer. A post-test is used to support the assumption that shoppers had expensive-price beliefs. Study 2 is a laboratory study that shows hypothesis 1b is one plausible explanation for the effects observed in the field. When consumers have expensive-price beliefs, a price delay increases purchase intent because consumers experience a more favorable price expectation disconfirmation. Study 3 is a field study that tests hypothesis 2a (i.e., a price delay decreases sales) using responses to a national retailer's email promotional appeals. A post-test is used to support the assumption that shoppers had inexpensive-price beliefs. Study 4 is a laboratory study that shows hypothesis 2b is one plausible explanation for the effects observed in the field. When consumers have inexpensive-price beliefs, a price delay decreases purchase intent because consumers experience a more unfavorable price expectation disconfirmation. Study 5 is a laboratory

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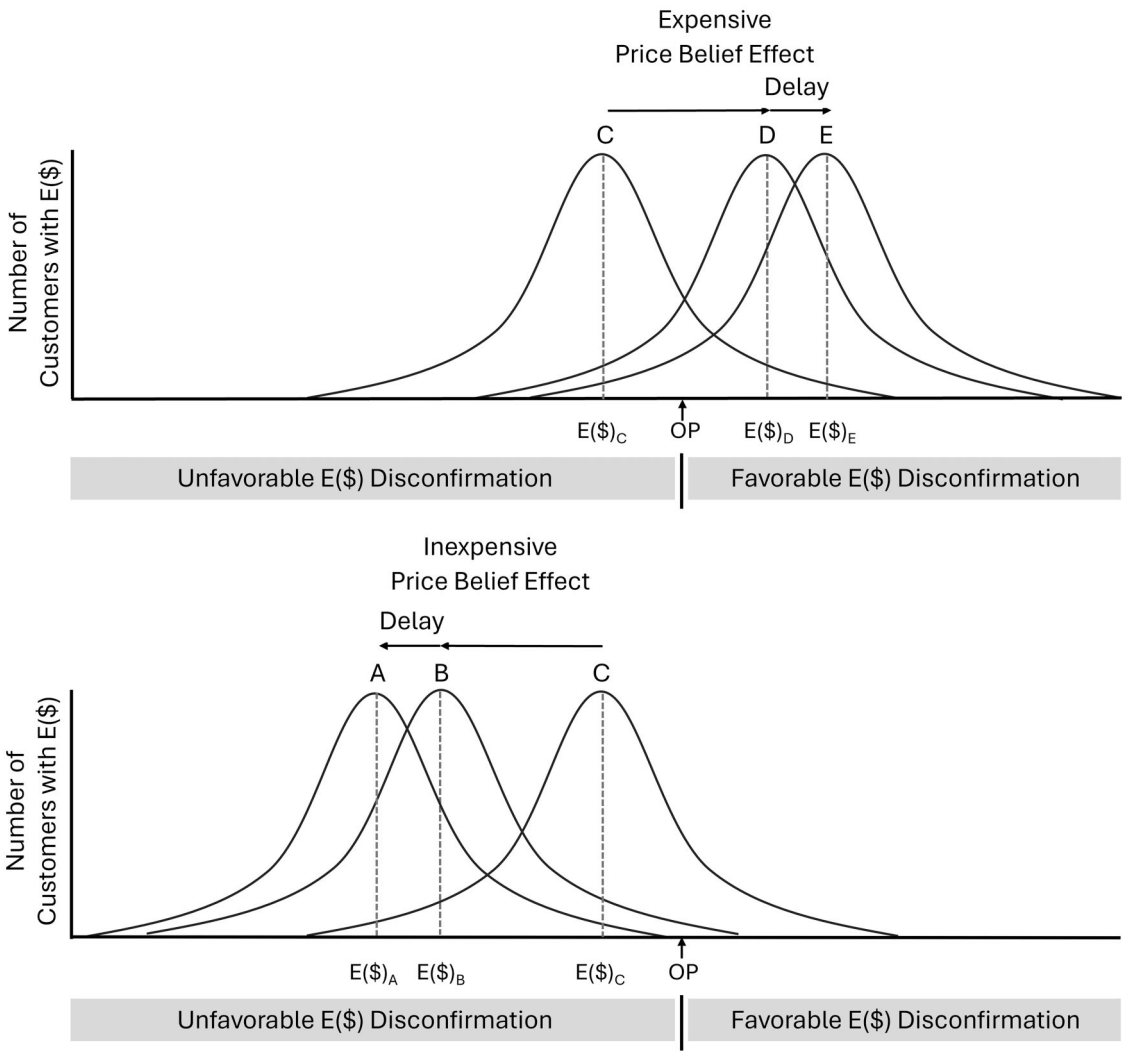
FIGURE 2

PRICE BELIEFS AND PRICE DELAY INFLUENCE PRICE EXPECTATIONS.



NOTE.— In the top panel of the figure, $E(\$)_C$ is the mean E($) of all target market consumers in the absence of price beliefs. $E(\$)_D$ is the mean E($) of all target market consumers in the presence of expensive-price beliefs. $E(\$)_E$ is the mean E($), of all target market consumers, in the presence of expensive-price beliefs and a price delay. Consumers with price expectations to the right (left) of the offer price (OP) should experience a favorable (unfavorable) price expectation disconfirmation and be more (less) likely to purchase the product. The bottom panel shows corresponding effects for target market consumers with inexpensive-price beliefs. In summary, when the dependent variable is purchase intention, it should increase (decrease) as favorable (unfavorable) price expectation disconfirmation becomes more extreme. E($), expected price.

study that manipulates price beliefs via store image and shows how a price delay increases (decreases) sales when people have expensive (inexpensive) price beliefs. Study 6 is a laboratory study showing how the disclosure of the reasons for a price delay moderates the influences of the price delay on expectation disconfirmation and purchase intention. Data and code for all studies are available for download at the Open Science Framework (OSF) repository (https://osf.io/xt42w/). The web appendix provides full materials and additional information for all studies.

STUDY 1: DELAYING PRICE IN AN ONLINE STORE INCREASES SALES

Study 1 tests hypothesis 1a—that when consumers believe a product will be expensive, a delayed price (vs. an

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immediate price) will increase sales. This field study was conducted in the online store of Fravega, an Argentinian retailer.[2] Fravega is a mid-tier retailer of household appliances and technology products, similar to Best Buy in the United States. Privately held Fravega had estimated sales of $2.3 billion in 2024 ($188 million in e-commerce). Fravega established its online channel in 2014 and routinely conducts up to five online A/B tests per month. This partnership provided us with a real-world setting to test our hypothesis.

In 2020, the year these data were collected, the inflation rate in Argentina was 42%. Thus, consumers' price beliefs associated with Fravega (i.e., mid-tier retailer), the product category and product line (see below), and the economic environment were likely expensive-price beliefs.

Method

Design and Data. The field study had two conditions: (1) immediate PD, with prices displayed alongside the products, and (2) delayed PD, with prices displayed only after a query was made (i.e., after a product was clicked on). In collaboration with the retailer, the experiment was conducted over an eight-week period (56 days) during the spring of 2020. The data and code are available on the OSF.

Procedure. Study 1 involved electric kettles (26 options) and espresso coffee machines (18 options). The procedure is illustrated in figure 3 using English language materials for espresso machines. Prospective customers who browsed either category saw a "thumbnail" for each product featuring an image and short description of each product. In the immediate PD condition, prices appeared in the thumbnails (see the four products on the left side of the display in figure 3). In the delayed PD condition, the pictures of the products and descriptions were identical, but the prices were omitted (see the four products in the center of the display in figure 3). If shoppers clicked on a product, in either condition, they were directed to the retailer's standard product page, which provided additional details on features and attributes, provided the price, and displayed an "add to cart" button (see the right side of the display in figure 3). Thus, all participants had to advance to the standard product page before they could add a product to their cart. Purchase behavior was recorded.

The retailer was responsible for implementing the A/B test. The retailer uses the VTEX platform to manage its e-commerce business. The VTEX platform supports A/B testing and can direct web traffic, at random, to one of the two experimental conditions. The platform has the following advantages: (1) random assignment to condition, (2) the ability to store experimental assignment information in browser cookies so a returning shopper using the same device can be reassigned to the same experimental condition, and (3) an ability to record customer behaviors during the shopping journey. A disadvantage of the platform is it uses simple random assignment (i.e., each participant who enters an A/B test has a 50/50 chance of being assigned to a condition); hence, there will be unequal sample sizes across conditions. A second disadvantage is it cannot retain experimental condition information if a consumer switches devices (e.g., the first visit was on a cell phone and the second visit was on a laptop).

Retailer Data Organization. The data were organized by day and experimental condition, collapsed over products and product categories. Thus, the data set consisted of sales metrics from two experimental conditions over eight weeks ($n = 112$). The retailer also recorded data on the volume of web visits. The average number of unique daily visitors was 267 to either of the product categories in the immediate PD condition and 280 to either of the product categories in the delayed PD condition. There was no information on whether any customers visited both product categories on a given day, but the retailer believed this was very unlikely.

Measures. In order to assess the robustness of the effect of the price delay manipulation, three unique shopping behaviors were analyzed. The first dependent measure was a *binary daily purchase indicator*, a useful metric for a retailer that operates in a high inflation environment. This measure assessed whether a sale occurred in either product category on that day. The binary daily purchase indicator was set to 1 if a purchase in either category took place on date t, or to 0 otherwise. The second measure was *daily units sold* (log transformed). Daily units sold refers to the total number of units purchased in an experimental condition on a given day. The third measure was *daily sales revenue* (log transformed). Daily sales revenue refers to the sum of the sale prices (in Argentinian pesos) for the units sold in a condition on a day. In addition, we calculated the bounce rate (the percentage of sessions ending after a single page view with no clicks to a standard product page) to check whether the delayed PD was more interesting to shoppers (i.e., more shoppers clicked through to standard product page).

Results

We estimate the model $DM_{i,\,t} = \alpha + \beta PD + \delta_t + \varepsilon_{i,\,t}$ for each dependent measure, where $DM_{i,\,t}$ denotes the dependent measure in experimental condition i on day t, PD is an indicator variable for the manipulation of PD equal to 0 in the immediate PD condition and 1 in the delayed PD condition, and δ_t denotes time fixed effects.[3] Fixed-time effects

2 Website: https://www.fravega.com/

3 There are 112 observations in the data (2 conditions × 56 days). With 55 control variables for days, 1 degree of freedom (df) for the constant, and 1 df for the treatment, there are therefore 55

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FIGURE 3

ILLUSTRATION OF CONDITIONS IN STUDY 1.

NOTE.—PD, price disclosure.

help control for the error associated with the high variability of a metric across days. Thus, the fixed-time effect reduces the error term without influencing the beta coefficient representing the effect of the price delay. Table 1 summarizes the results.

Given that the daily unit sales and daily revenue are log transformed, as required by their distribution, we report the statistics for these dependent measures using the model discussed above. However, we also report the means of conditions from the raw data to aid interpretability.

Binary Daily Purchase Indicator. Relative to the immediate PD, the delayed PD increased the percentage of days at least one customer made a purchase ($M_{\text{immediate}}$ = 0.804 or 45 days, M_{delayed} = 0.947 or 53 days, $\beta = 0.143$, standard error [SE] = 0.054, $t(55) = 2.66$, $p = .010$, $r^2 = 0.114$).

Daily Log Units Sold. Relative to the immediate PD, the delayed PD increased the daily log units sold ($M_{\text{immediate-log}} = 1.111$, $M_{\text{delayed-log}} = 1.259$; $\beta = 0.148$, SE = 0.082, $t(55) = 1.82$, $p = .075$, $r^2 = 0.057$; $M_{\text{immediate-raw}}$ = 2.911 units, $M_{\text{delayed-raw}}$ = 3.143 units).[4]

Daily Log Sales Revenue (Pesos). Relative to the immediate PD, the delayed PD increased the daily log sales revenue ($M_{\text{immediate-log}} = 7.806$, $M_{\text{delayed-log}} = 9.054$, $\beta = 1.248$, SE = 0.514, $t(55) = 2.43$, $p = .018$, $r^2 = 0.097$; $M_{\text{immediate-raw}}$ = 21,893.77 pesos, $M_{\text{delayed-raw}}$ = 29,109.71 pesos).[5,6]

dfs remaining for tests. Applying ordinary least squares to a binary outcome is a linear probability model (LPM). An LPM has implementation and interpretation advantages that make it more appealing than logistic regression. In LPM, parameters represent mean marginal effects that are easier to interpret than log odds ratios. Because LPM and logistic regression usually yield similar results (i.e., the impact of violating assumptions about the distribution of the dependent measure is minimal), econometric researchers tend to favor LPM for estimating treatment effects (Deke 2014).

4 We conducted a two-stage analysis decomposing units sold and revenue conditional on a sale being made on the day (i.e., the binary daily purchase indicator). The results showed the increased units sold (revenue) occurred because the price delay increased the likelihood of a sale being made on a day but not because there were more units sold (revenue) on days when there was a sale. See web appendix A.

5 The Argentinian peso means represent the experimental condition means before the data were log transformed. The official exchange rate on March 2, 2020, was 64 pesos to the dollar. The unofficial exchange rate was 83.86 pesos to the dollar. Thus, the peso amounts should be used to understand magnitudes of sales by condition rather than to assess dollar revenue to the firm during the test.

6 The delayed price condition (n = 280 per day) averaged 4.86% more visitors per day than the immediate price condition (n = 267). This could have impacted the dependent measures in a hypothesis-consistent direction.

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TABLE 1

ONLINE STORE STUDY—SUMMARY OF RESULTS

	Probability daily purchase	Log daily units	Log daily revenue	Bounce rate (%)
Delay prices	0.143**	0.1480†	1.248*	3.001**
	(0.054)	(0.082)	(0.514)	(0.560)
Constant	0.804**	1.111**	7.806**	20.868**
	(0.038)	(0.058)	(0.363)	(0.340)
Observations	112	112	112	112

NOTE.—The dependent variables are computed daily by experimental condition and collapsed across the two product categories. Units sold and sales revenue are log transformed. Standard errors are in parentheses.
†$p < .10$.
*$p < .05$.
**$p < .01$.

Daily Bounce Rate. The daily bounce rate was higher in the delayed PD condition (23.869%) than the immediate PD condition (20.868%; $\beta = 3.001$, SE $= 0.560$, $t(55) = 5.36$, $p < .001$, $r^2 = 0.344$). Note that higher bounce rates decrease, rather than increase, the likelihood that a sale will occur in the delayed PD condition, making the hypothesis test conservative.

Robustness Checks. In web appendix A, we report the estimation of a nonlinear model for the binary daily purchase indicator, which yields similar results.

Post-Test: Expensive-Price Beliefs. We recruited Prolific participants who were interested in electric kettles ($n = 78$) and espresso machines ($n = 79$) and tested the assumption that consumers had expensive-price beliefs in the shopping situation. We told participants, "In this scenario, you are browsing through a variety of electric kettles [espresso machines] on a store website. Your search leads to a website displaying multiple electric kettles [espresso machines]. The website shows pictures of the products and provides short descriptions, but there are no prices displayed—you have to click on the picture of the product to advance to a page that shows more information on the product, including the price. See below."

We used Fravega's website category pages for electric kettles and espresso machines to create thumbnails similar to those used in the delayed PD condition of the field study. Then, we translated the product information into English while keeping the exact layout used in the field study for ecological validity (web appendix A). We asked people, "Does the fact that you are not seeing the price lead you to believe that. . .," 1 = "The price is so low, they do not want to initially show it," to 7 = "The price is so high, they do not want to initially show it." We found that the mean score was higher than the midpoint for electric kettles ($M = 6.08$, $t(77) = 16.82$, $p < .001$) and espresso machines ($M = 6.29$, $t(78) = 21.15$, $p < .001$), suggesting that consumers in this setting likely held expensive-price beliefs. Our post-tests, in this and subsequent studies, validate that price delays increase the salience of directional price beliefs.

Alternative Explanations. It is possible that the withholding prices encouraged consumers to make favorable inferences about product quality or the retailer. Using a procedure similar to the post-test described above, we asked 75 (87) people interested in electric kettles (espresso machines) whether the hidden price decreased/increased perceived product quality, their trust in the retailer, and customer service expectations (see web appendix A for details). The mean scores for all measures in both categories did not differ from the midpoint (all $ps > .227$).

Discussion

Consistent with hypothesis 1a, study 1 provides evidence that delayed price exposure can increase the sales of electric water kettles and espresso machines. The process account of the effect cannot be determined because the field study was on an active retail website through which consumers could not be interviewed. Yet, there are three pieces of information that are consistent with customers having expensive-price beliefs. First, the retailer is a major department store chain that targets the middle class. Second, the data were collected in a high inflation period. Third, a post-test showed that a price delay resulted in people expecting higher prices than was the norm.

STUDY 2: DELAYING PRICE FOR A PREMIUM PRODUCT INCREASES PURCHASE INTENTIONS

Study 2 (preregistration: https://aspredicted.org/wj3cc2ws.pdf) is a laboratory study that conceptually replicates study 1. We sought to do a conceptual replication because we could not create a website with multiple stock-keeping units (SKUs), which manipulated the timing of prices. Instead, we created a situation that encouraged expensive price beliefs for a single product. The study used a premium espresso machine that was representative of the premium espresso machines used in study 1. We expected that delayed prices, relative to immediate prices, would increase purchase intentions (hypothesis 1a) and favorable

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price expectation disconfirmation (hypothesis 1b). We expected participants to have expensive-price beliefs in this shopping situation (confirmed in a post-test). Thus, study 2 could offer a plausible explanation for the price delay effect observed in study 1.

Method

Participants and Design. The study had two between-subjects PD conditions: immediate PD versus delayed PD. Following our preregistration, we initially recruited 300 Prolific participants who were then prescreened based on their interest in espresso machines. Specifically, we asked about people's interest in espresso machines on a five-point scale (1 = no interest at all, 2 = low interest, 3 = neutral, 4 = moderate interest, and 5 = a lot of interest). Those who answered 1, 2, or 3 were screened out and paid for their time. Those interested in espresso machines (4 or 5) continued the survey. Our final sample size had 229 participants (60.3% female, M_{age} = 37.38), who were randomly assigned to the conditions and paid a fixed nominal payment.

Procedure. Participants imagined they were interested in buying an espresso machine. In the scenario, participants imagined they were browsing through espresso machines on a store website and found an espresso machine that interested them. Participants in the immediate PD condition saw a picture of the espresso machine, information about attributes (brand, automation level, bars of pressure, integrated grinder, customer ratings, preheat time), and price, then added it to their shopping basket. Participants in the delayed PD condition saw the same espresso machine and information about attributes, but not the price, and then added it to their shopping basket, which is when they finally saw the price. The price ($142.50) was the median price in a pretest (see below for details). See web appendix B for stimuli.

Next, all participants indicated their likelihood of purchasing the espresso machine ("How likely would you be to buy this espresso machine?" 0 = extremely unlikely, 50 = somewhat likely, 100 = extremely likely) and price expectation disconfirmation ("Before seeing the actual price of the espresso machine. . .," 1 = "I expected a much lower price than its current price," to 4 = "I expected a price equal or similar to its current price," to 7 = "I expected a much higher price than its current price"). The purchase intention and expected price disconfirmation measures were counterbalanced. There were no main effects or interactions involving the dependent variable and mediator measurement order (web appendix B). Lastly, participants reported their gender and age.

Pretest. A pretest was used to set the price of the espresso machine and verify expensive-price beliefs for the espresso machine. Prolific participants (n = 82) interested in espresso machines were asked to look at the espresso machine and estimate its price in an open-ended response (M = $167.68, SD = 117.51). The median price ($142.50) was used to set the price in the study. Then, participants responded to a price belief question: "Relative to a typical espresso machine, I expected the price of this espresso machine to be. . ." (1 = much less expensive than a typical espresso machine, 50 = neither as inexpensive or expensive as a typical espresso machine, and 100 = much more expensive than a typical espresso machine). The mean (M = 58.79, SD = 19.38) was significantly above the midpoint of the scale ($t(81) = 4.11$, $p < .001$).

Results

Purchase Likelihood. We predicted that delayed PD would increase purchase likelihood. Consistent with our prediction, a one-way analysis of variance (ANOVA) revealed a higher likelihood of buying the espresso machine in the delayed PD condition (M = 62.80, SD = 24.60) than in the immediate PD condition (M = 51.86, SD = 27.97; $F(1, 227) = 9.88$, $p = .002$, $\eta_p^2 = 0.042$).

Price Expectation Disconfirmation. According to hypothesis 1b, delayed PD should make price expectation disconfirmation more favorable. Consistent with our prediction, a one-way ANOVA revealed a more favorable price expectation disconfirmation in the delayed PD condition ($M_{delayed} = 4.33$, SD = 1.60; $M_{immediate} = 3.75$, SD = 1.38; $F(1, 227) = 8.76$, $p = .003$, $\eta_p^2 = 0.037$).

Mediation. We ran a mediation analysis (PROCESS macro model 4; Hayes 2022) with 5,000 bootstrapped samples, using PD (0 = immediate, 1 = delayed) as the independent variable, the likelihood of purchasing the espresso machine as the dependent variable, and price expectation disconfirmation as the mediator. Consistent with hypothesis 1b, the indirect effect via price expectation disconfirmation ($\beta = 5.86$, SE = 1.98, 95% confidence interval [CI] = [2.00–9.81]) was significant.

Post-Test: Expensive-Price Beliefs. A post-test similar to that of study 1 was used to confirm delayed prices led to an expectation of higher prices. We showed 89 Prolific participants interested in espresso machines the same scenario and product from study 2's delayed PD condition. We told participants, "Imagine you are shopping online and interested in this espresso machine. You can't see the price yet—this happens in situations where the price tag is missing and you have to ask for more information, or when you have to add the product to the shopping cart to see the price. Does the fact that you are not seeing the price lead you to believe that. . .," 1 = "The price is so low, they do not want to initially show it," to 7 = "The price is so high, they do not want to initially show it." We found that the mean score was higher than the midpoint (M = 5.83, $t(88)$

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$= 16.82$, $p < .001$), suggesting that expensive-price beliefs influenced price expectations in the delayed PD condition.

Alternative Explanations. It is possible the withholding prices encouraged consumers to make favorable inferences about product quality or the retailer. Using a procedure similar to study 2's (condition: immediate PD vs. delayed PD), we asked 235 people interested in espresso machines about perceived product quality and customer service expectations (see web appendix B for details). There was no effect of condition on perceived product quality ($F(1, 233) = 0.915$, $p = .340$) and customer service expectations ($F(1, 233) = 0.354$, $p = .552$).

Discussion

Study 2 provided evidence for hypothesis 1a and hypothesis 1b. In a setting in which information about the product and the context of the purchase prompted consumers to believe prices would be expensive, delayed PD increased consumers' likelihood of buying the product. The effect was mediated by favorable price expectation disconfirmation, suggesting that the expensive-price belief exerted a stronger influence on expected prices when the price was delayed (see distributions D and E in figure 2).

STUDY 3: DELAYING PRICE IN A SALES FLYER REDUCES SALES

Study 3 tested hypothesis 2a—that when consumers have inexpensive-price beliefs, a delayed price (vs. an immediate price) will decrease sales. Study 3 was conducted in cooperation with Fravega in December 2021. In this instance, the retailer emailed a daily sales flyer (on seven consecutive days) to a database of over 700,000 current (high-potential customers in the customer database) and prospective customers. Similar to many email sales flyers, the flyer featured a limited selection of products (from 8 to 12 unique products each day, changing each day over the seven-day week, resulting in 70 unique advertised products) offered at a discount. To help communicate the opportunity for a deal, the first word of the email subject line included the word "Deals" or similar terminology. Thus, the price beliefs associated with the purchase opportunity were likely inexpensive. The influence of inexpensive-price beliefs on price expectations, when there are price delays, will be verified in a post-test.

Method

Procedure. The field study was conducted using a sales flyer that the retailer regularly emailed. Management had historical evidence that the flyer was a powerful tool for driving engagement and promoting transactions. On this occasion, the retailer emailed the sales flyer on seven consecutive days to 771,583 selected addresses in its database. The subject line of the seven email flyers included the word "Deals" or similar terminology, indicating a promotional shopping context. Each of the seven flyers advertised between 8 and 12 products from various categories (electric kettles, video consoles, headphones, microwaves, blenders, speakers, etc.), all offered at a promotional price. The emails contained thumbnail product pictures with a format similar to that used in study 1. Examples of the flyers are shown in web appendix C.

Email recipients were randomly assigned to the immediate PD condition, in which products and prices appeared together on the thumbnail in the sales flyer, or to the delayed PD condition, in which prices were omitted from the thumbnail in the sales flyer. In both conditions, a click on the thumbnail in the flyer sent the participant to the standard product page that provided details on features and attributes, listed the price, and displayed an "add to cart" button. In this experiment, 53.7% of consumers clicked on at least one product in at least one email, resulting in 4.096% of links clicked (i.e., on average, people clicked on 2.867 of the 70 advertised products).

Measures. The measures were similar to those used in study 1, except that they were (1) at the individual level, and (2) aggregated over the seven-day period. The *binary weekly purchase indicator* was set to 1 if a consumer made at least one purchase during the week, or to 0 if the consumer made no purchase during the week. Thus, it is the percentage of consumers who made a purchase. The *weekly units sold* (log transformed) measured how many total units were sold to a customer during the week. The *weekly sales revenue* (log transformed) measured the sales (in Argentinian pesos) revenue from the items purchased by a consumer during the week.

Data. The experiment was run during December 2021, and the data refer to email newsletters sent to 771,583 customers. The data and analyses code are posted on the OSF.

Results

We estimate the model $DM_i = \alpha + \beta\text{PD} + \varepsilon_i$, where DM_i denotes the dependent measure of consumer i, and PD is an indicator variable for the manipulation of PD equal to 0 in the immediate PD condition and 1 in the delayed PD condition. Table 2 summarizes the results. Precision is shown at four digits because the parameter estimates are small.

Binary Weekly Purchase Indicator. Relative to the immediate PD, the delayed PD decreased the percentage of customers who made a purchase during the week ($M_{\text{immediate}} = 2.6075\%$, $M_{\text{delayed}} = 2.3101\%$, $\beta = -0.2977$, $\text{SE} = 0.0352$, $t(771581) = -8.43$, $p < .001$, $r^2 = 0.0001$).

Weekly Log Units Sold. Relative to the immediate PD, the delayed PD decreased the weekly log units sold

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TABLE 2

EMAIL STUDY—SUMMARY OF RESULTS

	Probability (%) weekly purchase	Per-user log weekly units	Per-user log weekly revenue
Delay prices	−0.2974**	−0.0022**	−0.0306**
	(0.0352)	(0.0003)	(0.0035)
Constant	2.6075**	0.0186**	0.2549**
	(0.0257)	(0.0002)	(0.0025)
Observations	771,583	771,583	771,583

NOTE.—The dependent variables are computed weekly by participant (i.e., collapsed across all emails). Weekly units sold and weekly sales revenue are log transformed. Standard errors are in parentheses.
†$p < .10$.
*$p < .05$.
**$p < .01$.

($M_{\text{immediate-log}} = 0.0186$, $M_{\text{delayed-log}} = 0.0164$; $\beta = -0.0022$, SE $= 0.0003$, $t(771581) = -8.54$, $p < .001$, $r^2 = 0.0001$; $M_{\text{immediate-raw}} = 0.0277$ units, $M_{\text{delayed-raw}} = 0.0243$ units).

Weekly Log Sales Revenue. Relative to the immediate PD, the delayed PD decreased the daily log sales revenue ($M_{\text{immediate-log}} = 0.2549$, $M_{\text{delayed-log}} = -0.2243$, $\beta = -0.0306$, SE $= 0.0035$, $t(771581) = -8.84$, $p < .001$, $r^2 = 0.0001$; $M_{\text{immediate-raw}} = 900.86$ pesos, $M_{\text{delayed-raw}} = 768.31$ pesos).[7,8]

Click-Through Rate. One concern is that the click-through rate in the delayed PD condition may have been lower than in the immediate PD condition, and this explains the lower sales in the delayed PD condition. The click-through rate for the delayed PD condition ($M = 2.92$ out of 70 products) was actually higher than the click-through rate in immediate PD condition ($M = 2.81$ out of 70 products, $t(771{,}581) = -11.32$, $p < .001$, $r^2 = 0.0001$).

Robustness Checks. Robustness checks (reported in web appendix C) show comparable negative results when we estimate a nonlinear model. In addition, we find similar results when restricting the analysis to customers who clicked on the email at least once, which suggests that the findings are not driven by differences in the design of the sales flyers across conditions.

Post-Test: Inexpensive-Price Beliefs. A post-test was used to confirm the assumption that consumers had inexpensive-price beliefs in the sales flyer context. We created a sales flyer that mimicked the delayed PD sales flyer in terms of number of products (nine), claimed discounts (e.g., 45% off), financing (e.g., flexible payment plan), and layout (web appendix C). We recruited 100 Prolific participants and told them to imagine they received a sales flyer via email. We asked participants, "Does the fact that you are not seeing the price lead you to believe that. . .," 1 = "The price is so low, they do not want to initially show it," to 7 = "The price is so high, they do not want to initially show it." We found that the mean score was lower than the midpoint ($M = 3.29$, $t(99) = -3.29$, $p = .001$), suggesting that, on average, consumers held inexpensive-price beliefs in the sales flyer situation.

Alternative Explanations. It is possible that withholding prices encouraged consumers to make unfavorable inferences about product quality or the retailer. Using a procedure similar to the post-test, we asked 100 Prolific participants whether the hidden price decreased/increased perceived product quality, their trust in the retailer, and customer service expectations (web appendix C). The mean scores did not differ from the midpoint (all ps $> .363$).

Discussion

Consistent with hypothesis 2a, study 3 provides evidence that delayed price exposure can decrease the sales of products offered via an email sales flyer. Similar to study 1, the process account of the effect cannot be determined because the field study recorded sales on an active retail website through which consumers could not be interviewed. Yet, there is evidence consistent with a majority of the customers having an inexpensive-price belief. A post-test showed that people expected prices to be lower than the norm when they were not listed in a sales flyer. Still, the effects were much weaker than in study 1. One explanation may be that there was price belief heterogeneity, both across and within consumers. For example, if the positioning of the retailer, product categories, and economic environment encouraged expensive-price beliefs and the sales flyer encouraged inexpensive-price beliefs, then the relative weight of the competing beliefs will determine if an individual

7 The Argentinian peso means represent the experimental condition means before the data were log transformed. The official exchange rate on December 2, 2021, was 106.25 pesos to the dollar. The unofficial exchange rate was 175.81 pesos to the dollar.

8 Due to the high proportion of zero purchases (97.4% in control, 97.7% in treatment) and non-normal residual distributions, we conducted comprehensive alternative specifications including negative binomial, zero-inflated Poisson, hurdle models, gamma generalized linear model, and two-stage analyses to address potential distributional concerns. All specifications confirm consistent negative treatment effects. See web appendix C for complete details.

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consumer's price expectations increase or decrease when prices are delayed.

STUDY 4: DELAYING PRICE FOR A DISCOUNTED PRODUCT REDUCES PURCHASE INTENTIONS

Study 4 (preregistration: https://aspredicted.org/v87m-z58r.pdf) is a conceptual replication of study 3. We sought to do a conceptual replication because we could not mimic the email sales flyer procedure in the laboratory. Instead, we designed a procedure in which all participants were encouraged to have inexpensive-price beliefs. The study used an espresso machine that was representative of the basic espresso machines seen in the sales flyers of study 3. Participants were told the espresso machine was on sale. We expected that a delayed price, relative to an immediate price, would decrease purchase intentions (hypothesis 2a) because of an unfavorable price expectation disconfirmation (hypothesis 2b). We expected participants to have inexpensive-price beliefs (confirmed in a post-test). Thus, study 4 could offer a plausible explanation for the price delay effect observed in study 3.

Method

Participants and Design. The study had two between-subjects PD conditions: immediate PD versus delayed PD. Following our preregistration, we initially recruited 300 Prolific participants who were then prescreened with the same criterion used in study 2. Our final sample size had 223 participants (57.8% female, M_{age} = 36.30), which were randomly assigned to the conditions and paid a fixed nominal payment.

Procedure. Participants imagined they were interested in buying an espresso machine. In the scenario, participants were checking their email when they saw a subject line that caught their attention: "Best Deals on Espresso Machines: Up to 30% Off." Then, they opened the email and saw a sales flyer for an espresso machine. The sales flyer was labeled, "Starts today: 30% off select Breville espresso machines," with a picture of the espresso machine with a large "SALE: 30% OFF" red label on it. Below the sales flyer, participants in the immediate PD condition saw, "SALE $100.09 ~~REG. $142.99~~ + FREE SHIPPING." Participants in the delayed PD condition saw, "SALE $? (Click to see deal) ~~REG. $?~~ + FREE SHIPPING." Participants also read, "After some consideration, you decide to click the promotion to see more information about the product." The next page showed the espresso machine, information about attributes, and the price. Whereas those in the immediate PD condition knew the price, those in the delayed PD condition saw the price for the first time. The price ($100.09) was near the median price in a pretest (see below for details). See web appendix D for stimuli.

Next, all participants indicated their likelihood of purchasing the espresso machine ("How likely would you be to buy this espresso machine?" 0 = extremely unlikely, 50 = somewhat likely, 100 = extremely likely) and expected price disconfirmation: "Before seeing the actual price of the espresso machine…," 1 = "I expected a much lower price than its current price," to 4 = "I expected a price equal or similar to its current price," to 7 = "I expected a much higher price than its current price." The purchase intention and expected price disconfirmation measures were counterbalanced. There were no main effects or interactions involving the dependent variable and mediator measurement order (web appendix D). Lastly, participants reported their gender and age.

Pretest. A pretest was used to set the price of the espresso machine and verify inexpensive-price beliefs for the espresso machine. Prolific participants (n = 76) interested in espresso machines were asked to look at the espresso machine (which had a 30% off discount, just as in the main study) and estimate its final (i.e., after the 30% off discount) price in an open-ended response. The median price ($100.00) was used to set the price in the study ($100.09). We added $0.09 to the discounted price to make the regular price before the 30% off discount ($142.99) end in $0.99 ($100.09/0.7 = $142.99), which has greater ecological validity.

Then, participants responded to a price belief question: "Relative to a typical espresso machine, I expected the price of this espresso machine to be…" (1 = much less expensive than a typical espresso machine, 50 = neither as inexpensive nor expensive as a typical espresso machine, and 100 = much more expensive than a typical espresso machine). The mean (M = 44.03, SD = 18.53) was significantly below the midpoint of the scale ($t(75) = -2.81$, p = .006).

Results

Purchase Likelihood. We predicted delayed PD would decrease purchase likelihood. Consistent with our prediction, a one-way ANOVA revealed a lower likelihood of buying the espresso machine in the delayed PD condition (M = 51.71, SD = 24.52) than in the immediate PD condition (M = 63.17, SD = 21.06; $F(1, 221) = 14.02$, $p < .001$, $\eta_p^2 = 0.060$).

Price Expectation Disconfirmation. According to hypothesis 2b, delayed PD should make price expectation disconfirmation more unfavorable. Consistent with our prediction, a one-way ANOVA revealed a more unfavorable price expectation disconfirmation in the delayed PD condition ($M_{delayed}$ = 3.83, SD = 1.29; $M_{immediate}$ = 4.36, SD = 1.33; $F(1, 221) = 9.11$, $p = .003$, $\eta_p^2 = 0.040$).

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Mediation. We ran a mediation analysis (PROCESS model 4; Hayes 2022) with 5,000 bootstrapped samples, using PD (0 = immediate, 1 = delayed) as the independent variable, the likelihood of purchasing the espresso machine as the dependent variable, and price expectation disconfirmation as the mediator. Consistent with hypothesis 2b, the indirect effect via price expectation disconfirmation ($\beta = -3.34$, SE = 1.42, 95% CI = [−6.40 to −.95]) was significant.

Post-Test: Inexpensive-Price Beliefs. A post-test similar to those in prior studies was used to confirm delayed prices led to an expectation of lower prices. We showed 82 Prolific participants interested in espresso machines the same scenario from study 4's delayed PD condition. We then asked participants, "Imagine you receive an email advertising this espresso machine on sale. The e-mail advertising does not show the price—you have to click to see the deal. Does the fact that you are not seeing the price lead you to believe that. . ." (1 = "The price is so low, they do not want to initially show it," to 7 = "The price is so high, they do not want to initially show it"). We found that the mean score was lower than the midpoint ($M = 3.32$, $t(81) = -3.06$, $p = .003$), suggesting that inexpensive-price beliefs influenced price expectations in the delayed PD condition.

Alternative Explanations. It is possible that withholding prices encouraged consumers to make unfavorable inferences about product quality or the retailer. Using a procedure similar to study 4's (condition: immediate PD vs. delayed PD), we asked 256 people interested in espresso machines their perceived product quality and customer service expectations (see web appendix D for additional details). There was no effect of condition on perceived product quality ($F(1, 254) = 0.829$, $p = .364$) and customer service expectations ($F(1, 254) = 0.420$, $p = .517$).

Discussion

Study 4 provided evidence consistent with hypothesis 2a and hypothesis 2b. In a setting in which information about the product and the context of the purchase prompted consumers to believe prices would be inexpensive, delayed PD decreased consumers' intention to buy the product. The effect was mediated by unfavorable price expectation disconfirmation, suggesting that the inexpensive-price belief exerted a stronger influence on expected prices when the price was delayed (see distributions A and B in figure 2).

STUDY 5: DELAYING PRICES WITH DIFFERENT RETAILER PRICE IMAGE BELIEFS

Studies 1 through 4 used post-tests to provide evidence for the influence of price beliefs on price expectations and purchase behavior. In study 5, we manipulated price beliefs associated with store image while trying to keep other sources of price beliefs constant. There is robust literature showing that price beliefs related to store image create strong and consistent influences on price expectations (Hamilton and Chernev 2013). Thus, we manipulated whether prices were delayed or immediately displayed in a store known for high prices (i.e., a premium retailer) versus low prices (a discount retailer). We anticipated delayed PD at a premium (discount) retailer would result in increased (decreased) purchase intention owing to more favorable (unfavorable) price expectation disconfirmation, as hypothesized in hypothesis 1a and hypothesis 1b (hypothesis 2a and hypothesis 2b). We also included a national retailer condition that was meant to be comparable to the department store used in studies 1 and 3. Given that we have argued that the mid-tier price positioning of Fravega contributed to expensive-price beliefs in study 1, it was appropriate to investigate this claim by including a "national retailer" condition and showing support for hypothesis 1a and hypothesis 1b. These predictions were preregistered at https://aspredicted.org/gfsy-6shc.pdf.

Method

Participants and Design. Study 5 had a 2 (PD: immediate PD vs. delayed PD) × 3 (store image: premium vs. national vs. discount) between-subjects design. Following our preregistration, we initially recruited 900 participants who were then prescreened with the same criterion used in studies 2 and 4. Our final sample size had 726 Prolific participants (56.5% female, $M_{age} = 39.63$) who were randomly assigned to the conditions and paid a fixed nominal payment. Web appendix E provides details of the stimuli, including images of how the products were presented across conditions.

Procedure. The first step was to elicit an IRP from each participant. Specifically, we asked, "Considering the variety of espresso machines available on the market, how much do you estimate the average price for a typical espresso machine to be? Please base your estimate on your general knowledge and understanding of the market, without looking up any prices (there are no right or wrong answers!). Use numbers only." We also reminded participants of the difference between espresso machines and regular coffee makers.

In the second step, we described an online retailer that sold espresso machines. In the national retailer condition,

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participants read that *Whole Latte Love*, founded in 1997, was the largest online retailer of coffee and espresso equipment. Further information on size, staff, and customer service reinforced the national retailer image. Participants in the Premium retailer condition were told *Whole Latte Love* specialized in products for discerning aficionados. The retailer only sold machines that passed extensive quality control guidelines. Additional statements reinforced the premium retailer image. Participants in the discount retailer condition were told *Whole Latte Love* offered unbeatable deals and was committed to providing great coffee experiences for less. Additional statements reinforced the discount retailer image (web appendix E).

In the third and final step, participants shopped. We strove to make the product display and shopping experience as ecologically valid as possible. We displayed the 12 espresso machines randomly in a 4×3 grid. For each espresso machine, we provided a picture and information on the brand, automation level, bars of pressure, integrated grinder, customer ratings, and preheat time. Some attributes were negatively correlated so that choosing between options required making tradeoffs. We instructed participants to search for their favorite option and place it in the shopping cart.

We manipulated PD such that participants in the immediate PD condition saw the price of each espresso machine alongside the other attribute information and participants in the delayed PD condition only saw the caption, "Add to cart to see price." This procedure was similar to how customers shopped in study 1. Importantly, to ensure that prices were relevant to each participant but within a fixed range, we used the IRP of each participant to generate prices ranging from 85% to 115% (in 2.5% increments) of a given IRP, adding $0.99 in each case for realism. For example, a participant who believed that the average price of a typical espresso machine is $300 saw 12 options priced at $255.99 (85% of $300 plus $0.99), $263.49 (87.5% of $300 plus $0.99), $270.99 (90% of $300 plus $0.99), and so on up to $345.99 (115% of $300 plus $0.99).[9] Figure 4 reproduces 3 of the 12 espresso machines this participant would have encountered in the immediate PD (panel A) or delayed PD (panel B) condition. The prices shown correspond to 95%, 107.5%, and 110% of the $300 IRP, respectively.

Measures. The shopping cart displayed the espresso machine chosen by each participant and a subtotal with the price. At this moment, we asked, "If you were to purchase an espresso machine, how likely would you be to purchase the one you selected?" (1 = "not likely at all" to 7 = "extremely likely"). Next, we measured the price expectation disconfirmation: "Before seeing the actual price of the espresso machine. . .," 1 = "I expected a much lower price than its current price," to 4 = "I expected a price equal or similar to its current price," to 7 = "I expected a much higher price than its current price." Lastly, participants reported their gender and age.

9 Although the prices of the espresso machines varied from participant to participant depending on their IRP, each option had a fixed percentage relative to IRP—that is, one option from the set was always priced at 85% of the IRP, another option was always priced at 87.5% of the IRP, and so on. The presentation order was randomized.

Results

Purchase Likelihood. As predicted, a 3 (store image) × 2 (PD) ANOVA on purchase likelihood showed a significant interaction ($F(2, 720) = 13.40$, $p < .001$, $\eta_p^2 = 0.036$). As shown in figure 5, in the premium condition, participants in the delayed PD condition were more likely to purchase ($M = 6.08$, SD = 0.98) than their counterparts in the immediate PD condition ($M = 5.79$, SD = 0.78; $F(1, 720) = 6.66$, $p = .010$, $\eta_p^2 = 0.009$). In the national condition, participants who faced delayed PD were more likely to purchase ($M = 6.06$, SD = 0.93) than participants who faced immediate PD ($M = 5.72$, SD = 0.75; $F(1, 720) = 9.25$, $p = .002$, $\eta_p^2 = 0.013$). However, in the discount condition, participants in the delayed PD condition were less likely to purchase ($M = 5.40$, SD = 0.98) than their counterparts in the immediate PD condition ($M = 5.79$, SD = 0.78; $F(1, 720) = 12.34$, $p < .001$, $\eta_p^2 = 0.017$).

Price Expectation Disconfirmation. There was a store image by PD interaction ($F(2, 720) = 15.08$, $p < .001$, $\eta_p^2 = 0.040$). In the premium and national conditions, participants in the delayed PD condition provided more favorable price expectation disconfirmation scores (premium: $M = 4.83$, SD = 1.18; national: $M = 4.81$, SD = 1.27) than participants in the immediate PD condition (premium: $M = 4.35$, SD = 1.11; $F(1, 720) = 12.40$, $p < .001$, $\eta_p^2 = 0.017$; national: $M = 4.33$, SD = 0.98; $F(1, 720) = 12.22$, $p < .001$, $\eta_p^2 = 0.017$). In the discount condition, however, participants in the delayed PD reported less favorable price expectation disconfirmation scores ($M = 3.97$, SD = 0.89) than participants in the immediate PD condition ($M = 4.41$, SD = 0.89; $F(1, 720) = 10.37$, $p = .001$, $\eta_p^2 = 0.014$).

Mediation. Accordingly, we conducted a moderated mediation analysis (PROCESS model 8, 5,000 bootstrapped samples; Hayes 2022) with PD as the independent variable (0 = immediate PD, 1 = delayed PD), store image as the moderator, price expectation disconfirmation as the mediator, and purchase likelihood as the dependent variable. The index of moderated mediation is significant when comparing the discount condition to the premium (index = 0.17; 95% CI = [0.09–0.27]) and national (index = 0.17; 95% CI = [0.09–0.28]) conditions. In the premium condition, the pathway to purchase likelihood through price expectation disconfirmation is positive and significant ($\beta = 0.09$, SE = 0.03, 95% CI = [0.03–0.15]). In the national condition, the same pathway is also positive and significant

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FIGURE 4

MANIPULATION OF PRICE DISCLOSURE

NOTE.—(A) Immediate price disclosure condition. (B) Delayed price disclosure condition.

($\beta = 0.09$, SE = 0.03, 95% CI = [0.03–0.15]). In the discount condition, however, the pathway to purchase likelihood through expectation disconfirmation is negative and significant ($\beta = -0.08$, SE = 0.03, 95% CI = [−0.14 to −0.04]). This pattern indicates that participants expected a higher (lower) price for their chosen option in the premium and national (discount) store image condition(s), which in turn increased (decreased) purchase likelihood.

Evidence for Assumptions About Expected Prices. The process explanation for the delayed price effect is that price beliefs shift expected prices to a greater extent when prices are delayed, and this results in greater price expectation disconfirmation once the price is revealed. Study 5 measured price expectation disconfirmation (mediator), but it did not measure the expected price. The reason for this decision relates to the immediate PD condition—it is

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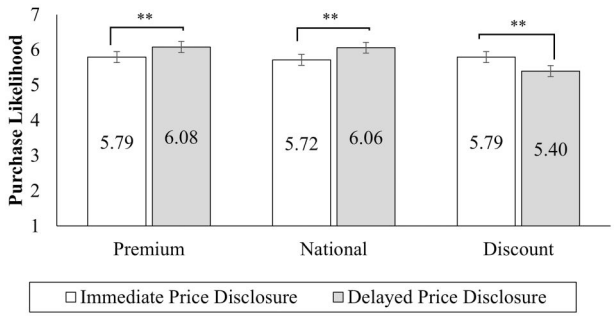
FIGURE 5

PURCHASE LIKELIHOOD ACROSS PRICE DISCLOSURE AND STORE IMAGE CONDITIONS



NOTE.—Error bars = ±1 standard error. $^{**}p < .01$.

unnatural for a participant to report an expected price on a product after its price is disclosed. To address this issue, we repeated study 5 with two changes. First, we measured price expectations about the product category rather than the product the participant selected. This should address the awkwardness in the immediate price condition. Second, we measured category price expectations (henceforth category IRP) at the beginning of the study (e.g., "Considering the variety of espresso machines available on the market, how much do you estimate the price for a typical espresso machine to be? __") and after the product had been placed in the shopping cart but before the price had been disclosed in the delayed PD condition. Detailed information is in web appendix F.

The crucial dependent measure was the second category IRP minus the first category IRP. As expected, an ANOVA on the category IRP differential found a significant interaction between store information and PD ($F(2, 591) = 15.46$, $p < .001$, $\eta_p^2 = 0.050$). In the premium and national conditions, the category IRP differential is positive and larger for participants in the delayed PD condition ($M_{\text{premium}} =$ \$67.91, SD = \$159.97; $M_{\text{national}} =$ \$61.94, SD = \$144.69) than for participants in the immediate PD ($M_{\text{premium}} =$ \$2.63, SD = \$36.03; $F(1, 591) = 21.83$, $p < .001$, $\eta_p^2 = 0.036$; $M_{\text{national}} =$ \$5.73, SD = \$24.76; $F(1, 591) = 16.19$, $p < .001$, $\eta_p^2 = 0.027$). Conversely, in the discount condition, the category IRP differential is negative and larger for participants in the delayed PD condition ($M = -$\$26.96, SD = \$99.17) than for participants in the immediate PD condition ($M =$ \$7.61, SD = \$17.25; $F(1, 591) = 6.03$, $p = .014$, $\eta_p^2 = 0.010$). Thus, price beliefs did influence category price expectations in the delayed PD condition.

Alternative Explanations. An analysis using the price of the chosen option found no main effect of PD ($F(1, 720) = 0.156$, $p = .693$), a marginally significant effect of store information ($F(2, 720) = 2.368$, $p = .094$) and, most importantly, no interaction effect ($F(1, 720) = 0.628$, $p = .534$). This is evidence against the alternative hypothesis that purchase likelihood was influenced by participants in the immediate price condition selecting higher (in premium and national condition) or lower (in discount condition) price-point products than people in the delayed price condition. In addition, we reran the same procedure from study 5 ($N = 776$ valid responses), and once participants had the product in their shopping cart, we asked the same product quality ($\alpha = 0.92$) and customer service expectations ($\alpha = 0.88$) questions used in studies 2's and 4's alternative explanation post-tests (web appendix F). There were no main effects or interaction effect on perceived product quality (Fs < 1). In addition, there was no main effect of PD or an interaction with store image (Fs < 1). There was, not unexpectedly, a main effect of store image ($F(2, 770) = 3.34$, $p = .036$, $\eta_p^2 = 0.009$), such that expectations of customer service were lower in the discount condition ($M = 6.07$, SD = 1.09) than in the national ($M = 6.28$, SD = 0.88; $F(1, 770) = 6.22$, $p = .012$, $\eta_p^2 = 0.008$) and premium ($M = 6.23$, SD = 0.96; $F(1, 770) = 3.25$, $p = .074$, $\eta_p^2 = 0.004$) conditions. Thus, our effects appear to be driven by price expectations rather than changes in perceptions about the product and/or the retailer.

Discussion

Overall, study 5 provides evidence that, relative to immediate PD, delayed PD amplifies price expectations, generating expectation disconfirmations that then influence purchase intentions. Web appendix G reports a successful replication in which participants considered purchasing a Bluetooth keyboard, store image had only two conditions (premium and discount), and prices were predetermined rather than a function of IRPs.

This study (and its conceptual replication) manipulated price beliefs through store image. However, there are additional ways to manipulate price beliefs that could have meaningful consequences for retailers. For example, when consumers have expensive-price beliefs, disclosing that a price is hidden because it is a bargain can encourage a shift to inexpensive-price beliefs. A practical, e-commerce example is when a retailer discloses a price is hidden because it violates an MAP policy (i.e., an agreement between a manufacturer and a retailer that stipulates the lowest price that can be advertised for a product). We investigated this practical moderator in study 6.

STUDY 6: THE MODERATING ROLE OF MAP POLICIES

An MAP policy is an agreement between a manufacturer and a retailer that stipulates the lowest price that can be advertised for a product. Retailers retain the right to sell at

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a price below the MAP, but they can only reveal it once consumers advance to the shopping cart—wherein the price is considered part of the sales agreement rather than a promotional hook. Consequently, many retailers that delay PD because of an MAP policy prompt consumers to "Add to cart to see the price." Whereas some US retailers in this situation simply delay PD, other retailers, including Academy Sports + Outdoors and Walmart, explain, "Price is too low to show," and attribute it directly to the MAP policy, providing reasons for the price delay (see web appendix H for examples).

The goal of this experiment was to test the possibility that attributing delayed PD to an MAP policy moderates its effect on purchase intentions. When consumers have expensive-price beliefs, an assumption that will be confirmed in a post-test to be discussed later, a price delay increases purchase intention (hypothesis 1a) owing to a favorable price expectation disconfirmation (hypothesis 1b). Yet, attributing the delayed disclosure to an MAP policy may signal to consumers that the price is likely low, effectively replacing an expensive-price belief with an inexpensive-price belief. When consumers have inexpensive-price beliefs, a price delay decreases purchase intention (hypothesis 2a) owing to an unfavorable price expectation disconfirmation (hypothesis 2b). Thus, attributing a price delay to an MAP policy can make the beneficial effects of a price delay (when the MAP policy is not disclosed) become detrimental (when the MAP policy is disclosed).

Method

Participants and Design. We recruited 322 Prolific participants. We randomly assigned participants to one of three PD conditions (immediate PD, delayed PD, and delayed PD with MAP policy). In line with the preregistration (https://aspredicted.org/kz3n-gzzd.pdf), we removed 22 participants who failed the attention check, resulting in an effective sample of 300 (45.7% female, $M_{age} = 38.36$).

Procedure. We explained, "In this survey, we want you to imagine you want to purchase a robot vacuum cleaner. In this scenario, you are browsing through a variety of robot vacuum cleaners on a website. At some point, you found a robot vacuum cleaner that interested you. Click the arrow button to continue." Web appendix H provides details of the stimuli.

On the next page, participants previewed the product "iRobot Roomba Combo i5+ Self-Emptying Robot Vacuum & Mop," which included the number of reviews and shipping date. Here, we randomly assigned participants to one of three PD conditions. Participants in the immediate PD condition saw the product's price ($349.99) and an "Add to cart" button. In this study, the product's price was based on the real product's price at the time the experiment was run rather than a pretest. Participants in the delayed PD condition did not see the price, and the button read, "See price in cart." Finally, participants in the delayed PD with MAP policy did not see its price, saw the "See price in cart" button, and saw a mouseover element in the shape of a question mark that read, "Please add this item to your cart to see the price. Because the price is below the manufacturer's minimum advertised price, we are unable to show it here." We then told all participants, "You decided to see more information about the product. Click the Next button to continue." On the following page, we provided additional product information. This information, which included a highlight of the main features, a one-paragraph description, and zoomed product pictures, was taken directly from Best Buy. Participants in the two delayed PD conditions still did not receive the price. After processing the additional details, participants returned to the product previews and read, "After some consideration, you decide to add the product to your shopping cart. Please click the next button to see the product in your shopping cart." Then, on the final page, participants viewed the shopping cart containing the product, its price, and the cart subtotal. This was the first time that participants in the delayed PD conditions saw the price.

Below the shopping cart, we asked, "If you were in this scenario, how likely would you be to purchase the option in your shopping cart (i.e., checkout)?" (1 = "not likely at all" to 7 = "extremely likely"). To measure price expectation disconfirmation, we then asked, "Consider your shopping experience, the product, and its price ($349.99). How would you describe your price expectations relative to its current price?" (1 = "I expected a much lower price than its current price," to 4 = "I expected a price equal or similar to its current price," to 7 = "I expected a much higher price than its current price").

Results

Purchase Likelihood. First, a three-cell ANOVA on purchase likelihood shows a main effect of PD ($F(2, 297) = 17.07$, $p < .001$, $\eta_p^2 = 0.103$). Specifically, participants were more likely to buy the vacuum cleaner in the delayed PD condition ($M = 5.01$, SD = 1.27) than in the immediate PD ($M = 4.43$, SD = 1.70; $F(1, 297) = 6.36$, $p = .012$, $\eta_p^2 = 0.021$) or delayed PD MAP ($M = 3.67$, SD = 1.85; $F(1, 297) = 33.94$, $p < .001$, $\eta_p^2 = 0.103$) conditions (figure 6). In contrast, participants were less likely to buy the vacuum cleaner in the delayed PD MAP condition than in the immediate PD condition ($F(1, 297) = 10.92$, $p < .001$, $\eta_p^2 = 0.035$).

Price Expectation Disconfirmation. Second, a similar ANOVA on price expectation disconfirmation reveals a main effect of PD ($F(2, 297) = 18.76$, $p < .001$, $\eta_p^2 = 0.112$). Specifically, participants reported higher scores in the delayed PD condition ($M = 4.52$, SD = 1.34) than in the immediate PD ($M = 4.02$, SD = 1.20; $F(1, 297) =$

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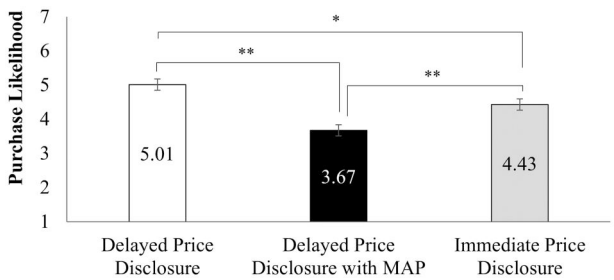
FIGURE 6

PURCHASE LIKELIHOOD ACROSS PRICE DISCLOSURE CONDITIONS



NOTE.—Error bars = ±1 standard error. MAP, minimum advertised price. $^{*}p < .05$; $^{**}p < .01$.

6.93, $p = .009$, $\eta_p^2 = 0.023$) or delayed PD MAP ($M = 3.36$, SD = 1.47; $F(1, 297) = 37.28$, $p < .001$, $\eta_p^2 = 0.112$) conditions. In contrast, participants reported lower scores in the delayed PD MAP than in the immediate PD condition ($F(1, 297) = 12.07$, $p < .001$, $\eta_p^2 = 0.039$).

Mediation. Accordingly, we conducted a mediation analysis (PROCESS model 4, 5,000 bootstrapped samples; Hayes 2022) with PD as the independent variable (multicategorical), expectation disconfirmation as the mediator, and purchase likelihood as the dependent variable. The pathway to purchase likelihood through expectation disconfirmation is positive and significant when contrasting the delayed PD condition to the immediate PD ($\beta = 0.23$, SE = 0.09, 95% CI = [0.06–0.42]) and delayed PD MAP ($\beta = 0.54$, SE = 0.12, 95% CI = [0.32–0.79]) conditions. In addition, the same pathway is negative and significant when contrasting the delayed PD MAP and immediate PD ($\beta = -0.31$, SE = 0.10, 95% CI = [−0.53 to −0.12]) conditions.

Post-Test: Price Beliefs. A post-test similar to those used in prior studies was used to confirm that delayed prices (vs. delayed prices with MAP policy attribution) led to an expectation of higher (lower) prices. We showed 79 (vs. 80) Prolific participants interested in robot vacuum cleaners the same scenario from study 6's delayed PD (vs. delayed PD with MAP attribution). We then asked participants, "Imagine you are shopping online and interested in this robot vacuum cleaner. You can't see the price yet—this happens in situations where the price tag is missing and you have to ask for more information, or when you have to add the product to the shopping cart to see the price. Does the fact that you are not seeing the price lead you to believe that…" (1 = "The price is so low, they do not want to initially show it," to 7 = "The price is so high, they do not want to initially show it"). We found that in the delayed PD condition, the mean score was higher than the midpoint ($M = 5.35$, $t(78) = 7.97$, $p < .001$), suggesting that expensive-price beliefs influenced price expectations in this condition. Likewise, in the delayed PD with MAP condition, the mean score was lower than the midpoint ($M = 3.29$, $t(79) = -3.12$, $p = .003$), suggesting that inexpensive-price beliefs influenced price expectations in this condition. Web appendix H provides additional details.

Alternative Explanations. We reran the same procedure from study 6 ($N = 300$ valid responses), and once participants had the robot vacuum cleaner in their shopping cart, we asked the same product quality ($\alpha = 0.95$) and expectations about customer service ($\alpha = 0.88$) questions used in previous post-tests. There was no effect of PD on perceived product quality ($F < 1$) or expectations about customer service ($F < 1$). Thus, our effects appear to be driven by price expectations rather than changes in perceptions about the product and/or the retailer. See web appendix H for details.

Discussion

The results support our intuition that attributing delayed PD to an MAP policy can moderate the effect of the tactic. Specifically, relative to the immediate PD and delayed PD conditions, the delayed PD MAP condition prompted participants to expect a low price, which resulted in disappointment once the price was revealed and reduced purchase intent.

GENERAL DISCUSSION

Withholding prices has long been a staple tactic in retail, yet its effectiveness remains a subject of enduring debate among businesses. Surprisingly, despite its prevalence, academic research offers limited empirical guidance to resolve these conflicting perspectives. This article addresses this long-standing disconnect. A central finding of this research is that delayed PD exerts a significant influence on consumer purchase behavior. However, the direction of this influence—whether it increases or decreases purchase likelihood—is contingent upon consumers' pre-existing price beliefs and their influence on price expectations. Holding back prices exacerbates the influence of price beliefs on price expectations (figure 2), increasing the likelihood that shoppers experience a positive (if they have expensive-price beliefs) or negative (if they have inexpensive-price beliefs) disconfirmation once prices are revealed. This disconfirmation then carries through to purchase decisions.

Contributions

First, our research provides guidance to retailers questioning or exploring the right moment to post a price. We show that a determining factor is the relationship between

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price beliefs and price expectations. By understanding this relationship, retailers can assess factors influencing price beliefs (e.g., store image, product category, economic conditions) and predict whether a price delay will boost or hinder sales.

Second, our research extends and enriches the established literature on price obfuscation in economics and marketing. This body of work has predominantly focused on firms' strategies to complicate price discovery by advertising a low base price and concealing supplementary charges for add-ons, additional services, taxes, and related expenses. In such scenarios, it is plausible to assume that some consumers will opt to minimize search costs or commit computational errors, leading to the general finding that price obfuscation tends to inflate the final sales price. In contrast, our investigation examines the more fundamental phenomenon of shrouding the entire product price. When consumers are temporarily prevented from accessing any price information, the arguments centered on search costs and computational errors become less compelling. Instead, we propose and test an alternative explanatory framework for the impact of price obfuscation, one that yields a more nuanced understanding of when withholding prices proves advantageous versus detrimental. Specifically, we demonstrate that consumers' pre-existing price beliefs shape their price expectations, which in turn influence their purchase decisions through the mechanism of price expectation disconfirmation.

Finally, our research offers valuable insights into the broader managerial debate surrounding the role of friction, or perceived hassle, in shopping experiences. Conventional wisdom often dictates the pursuit of frictionless purchase journeys, particularly when it comes to pricing and payment processes. However, our studies demonstrate that not all friction is inherently detrimental. Indeed, it may seem counterintuitive that a retailer could strategically benefit from introducing friction, yet this is precisely what delayed PD accomplishes. This finding suggests a promising avenue for companies to explore: the implementation of an "adaptive" approach to PD, wherein the timing of price presentation is personalized based on individual shoppers' price beliefs. Such an approach could strike a balance between enhancing the shopping experience and achieving sales outcomes—an optimal balance between friction and flow in the purchase journey (Bertini, Aparicio, and Aydinli 2024).

Limitations and Future Research

To the best of our knowledge, this article reports the first empirical test of the effect of delayed PD on consumer behavior. In this initial investigation, we do not address heterogeneity in price beliefs among consumers. Heterogeneity occurs when one portion of a consumer population has expensive-price beliefs and another portion of the consumer population has inexpensive-price beliefs. The relative size of these two groups determines the degree and direction of the price expectation shift, and corresponding impact on sales, when there is price delay. This may be why the positive effect of the price delay was strong in study 1 (i.e., a large majority of customers had expensive-price beliefs) and the negative effect of the price delay was weak in study 3 (i.e., a small majority of customers had inexpensive-price beliefs). Identifying indicators of the direction and strength of price beliefs, at the individual customer level, would be a useful tool for price discrimination.

Moreover, although our research emphasizes the role of price expectations as a key explanatory mechanism, we acknowledge that this is likely one of several contributing processes. To illustrate, consider research showing that price promotions can cause reward-seeking and impatience (Shaddy and Lee 2020), making consumers more present oriented. Consumers who are impatient might find delayed PD frustrating, as it prevents them from quickly obtaining all the information they need to decide, leading to negative affect and/or abandonment of the purchase process. Although this explanation is admittedly simplified, it offers a plausible alternative account for the effects observed in study 3. Future research could investigate this and other possibilities.

Beyond the role of price expectations, another critical moderator to consider is price importance—the relative weight consumers assign to price compared to other product benefits. When price is deemed less important in a purchase decision, delaying its disclosure may paradoxically enhance product appeal. In such instances, consumers are likely to focus more intently on the immediately available product features and advantages. Conversely, when price is a primary driver of the decision, delaying its disclosure could shift attention disproportionately to the missing information, potentially diminishing the perceived value of other product benefits. Consequently, understanding how and when price importance moderates the impact of the price delay effect represents a logical next step in this research stream.

Our studies adopt a price delay duration that we think is meaningful to consumers but conservative given that it extends only to the next logical step in the purchase process. Longer price delay durations are certainly possible, all the way to the moment of payment. Varying the duration of the price delay is an interesting extension of our work, one that would allow researchers to contrast the possibility of an escalation of beneficial shifts in price expectations with the risk of making the shopping experience less realistic or less natural to consumers.

Another promising avenue for future research lies in exploring whether delayed PD can function as a behavioral intervention, empowering consumers to select products based on genuine preferences rather than mere cost justification. By withholding the price, consumers may be

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prompted to focus more intently on the product's inherent features and benefits, thereby aligning their decisions more closely with their true desires rather than being predominantly driven by price considerations. This perspective expands upon the notion of positive expectation disconfirmation, suggesting that delayed PD has the potential to enhance the congruence between consumer preferences and their purchase decisions.

Furthermore, it would be interesting to extend our research to consider competition. If prices are disclosed not only to consumers but also to competitors, then is it better for a retailer to hide them from the more "public" places of the purchase process (a store shelf, a landing page, etc.)? The answer to this question probably depends on several factors, including the underlying pricing policy of the retailer and its competitors. For instance, a company with a policy of matching the lowest price might benefit from posting prices immediately because consumers who compare prices will have less friction in the shopping experience.

Finally, identifying potential moderators of the price delay effect warrants further investigation. Individual differences, such as dispositional optimism or pessimism, could influence a consumer's propensity to utilize price beliefs when updating price expectations. Similarly, price knowledge and confidence may mitigate the effects of delayed PD, as more knowledgeable consumers are likely less susceptible to revising their expectations during the delay. This implies price delay effects are less likely to manifest for fast-moving consumer goods, wherein consumers possess extensive price histories. Additionally, when a price delay is perceived as non-normative—deviating from standard practice in a given retail context—consumers may be less inclined to allow their price beliefs to sway their expectations, potentially attenuating the effects observed in our studies.

DATA COLLECTION STATEMENT

Data from study 1 were provided by Fravega, an Argentinian retailer of household and tech appliances, to the third author. Collected in spring 2020 by the company under the third author's supervision, these data were analyzed by the third author with input from the first and sixth authors. The post-test and alternative explanations tests for study 1 were conducted in February 2025 by the first author, who also analyzed the data. Study 2's main study and pretest were conducted via Prolific in February 2025 by the first and second authors. The post-test and alternative explanations tests were conducted online in March 2025 by the same authors. All study 2 data were analyzed by the first and second authors. Data for study 3 were again provided by Fravega to the third author. Collected in December 2021 under the third author's supervision, these data were analyzed by the third author with input from the first and sixth authors. The post-test and alternative explanations tests for study 3 were conducted in February 2025 by the first author, who also analyzed the data. Study 4's main study and pretest were conducted via Prolific in February 2025 by the first and second authors. The post-test and alternative explanations tests were conducted in March 2025 by the same authors, who analyzed all tests and study data. Study 5's main study and alternative explanations test were conducted via Prolific in March 2025 by the first author and analyzed by the first and second authors. The web appendix F study (i.e., study 5 procedure with category-level price expectations) was conducted via Prolific in April 2024 by the first author and analyzed by the first and second authors. The web appendix G study (a conceptual replication of study 5) was also conducted via Prolific in April 2024 by the first author and analyzed by the first and second authors. Data for study 6 were collected via Prolific in August 2024 by the first author and analyzed by the first and second authors with input from all other authors. The post-test and alternative explanations tests for study 6 were conducted in March 2025 by the first author and analyzed by the first and second authors. The raw data for all studies, code (SPSS for laboratory studies and STATA/R for field studies), qualtrics survey files (.qsf), and preregistration documents for the laboratory surveys and studies are available on the OSF website (https://osf.io/xt42w/).

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